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(2)

Measures of Central Tendency

Correcting and Combining the Arithmetic Mean

Measures of Central Tendency (The Mean)

Correcting the Arithmetic mean

Example 1:- The arithmetic mean of 75 observations is 46. Later it was found that the two numbers 66 and 85 were misread as 36 and 45. Find the correct arithmetic mean.

Measures of Central Tendency (The Mean)

Example 2:- The average mark in Calculus I of 70 students is 77. Later it was found that the two marks of two students which are 58 and 81 were left out at the time of calculation. Find the average mark of the 72 students.

Measures of Central Tendency (The Mean)

Combining the Arithmetic mean

Example 1:- The arithmetic mean of 40 observations is 27 and the arithmetic mean of another 55 observations is 64. Find the arithmetic mean of the 95 observations together.

Measures of Central Tendency (The Mean)

Solution:- Given $\bar{x}_{40} = 27$ and $\bar{x}_{55} = 64$

$$\text{Since } \bar{x}_{40} = \frac{\sum_{i=1}^{40} x_i}{40} \Rightarrow \sum_{i=1}^{40} x_i = \bar{x}_{40} \times 40$$

$$\Rightarrow \sum_{i=1}^{40} x_i = 27 \times 40 = 1080$$

$$\text{and since } \bar{x}_{55} = \frac{\sum_{i=1}^{55} x_i}{55} \Rightarrow \sum_{i=1}^{55} x_i = \bar{x}_{55} \times 64$$

Measures of Central Tendency (The Mean)

$$\Rightarrow \sum_{i=1}^{55} x_i = 55 \times 64 = 3520$$

$$\text{Thus } \bar{x}_{40+55} = \bar{x}_{95} = \frac{\sum_{i=1}^{40} x_i + \sum_{i=1}^{55} x_i}{40+55} =$$

$$\frac{\sum_{i=1}^{95} x_i}{95} = \frac{1080 + 3520}{95} = \frac{4600}{95} = 48.42$$

In general $\bar{x}_C =$

$$\frac{\sum_{i=1}^{N_1} x_1 + \sum_{i=1}^{N_2} x_2 + \dots + \sum_{i=1}^{N_k} x_k}{N_1 + N_2 + \dots + N_k}$$

Measures of Central Tendency (The Mean)

Example 2:- The average daily wage paid to all male employees in the management group of a company is 255 birr, while the average daily wage of female management group employees is 345 birr. If there are 60 employees in a company's management group and the average daily wage paid to all employees in this management group is 307.50 birr, then find the number of male and female workers of the management group employed in a company.

Measures of Central Tendency (The Mean)

Merits and demerits AM

Merits:

- Based on all observations
- Easy to calculate and simple to understand
- Widely used in statistical analysis
- It can be used for comparison and for further mathematical conclusion

Measures of Central Tendency (The Mean)

Demerits:

- Can not be computed for qualitative data.
- It is highly affected by extreme values (either low or high).
- Sometimes gives absurd results (4.4 children).
- If a single observation is missing or lost arithmetic mean cannot be calculated.

The Geometric Mean

Measures of Central Tendency (The Geometric Mean)

The geometric mean provides an *accurate measure of the average percentage rate of a series of numbers through time.*

Definition:- The geometric mean, denoted by GM is found by taking the n^{th} root of the product of the given n numbers.

$$\text{i.e. GM} = \sqrt[n]{x_1 \times x_2 \times x_3 \dots \times x_n}$$

Example 1:- Find the AM and GM of 5, 6, 8, 9, 11

Measures of Central Tendency (The Geometric Mean)

Solution1:- We know that $AM = \mu(x) = \frac{5 + 6 + 8 + 9 + 11}{5}$

$$= \frac{39}{5} = \mathbf{7.8}$$

Since $GM = \sqrt[n]{x_1 \cdot x_2 \cdot x_3 \cdot \dots \cdot x_n}$, it follows that

$$Gm = \sqrt[5]{5 \cdot 6 \cdot 8 \cdot 9 \cdot 11} = \sqrt[5]{23760} = \mathbf{7.5019}$$

As we can see $GM < AM$ and in general $GM \leq AM$.

$$\text{Since } GM = \sqrt[n]{x_1 \cdot x_2 \cdot x_3 \cdot \dots \cdot x_n} =$$
$$\left(x_1 \cdot x_2 \cdot x_3 \cdot \dots \cdot x_n \right)^{1/n}$$

$$\Rightarrow \log GM = \log \left(x_1 \cdot x_2 \cdot x_3 \cdot \dots \cdot x_n \right)^{1/n}$$

Measures of Central Tendency (The Geometric Mean)

$$= \frac{1}{n} [\log (x_1 \cdot x_2 \cdot x_3 \cdot \dots \cdot x_n)]$$

$$= \frac{1}{n} [\log x_1 + \log x_2 + \log x_3 + \dots + \log x_n]$$

$$\Rightarrow \text{GM} = \text{Antilog } \frac{1}{n} [\log x_1 + \log x_2 + \log x_3 + \dots + \log x_n]$$

Example 2:- Consider example 1 and use the logarithm formula to find the GM.

Solution2:- Given 5, 6, 8, 9, 11

Measures of Central Tendency (The Geometric Mean)

$$= \frac{1}{n} [\log (x_1 \cdot x_2 \cdot x_3 \cdot \dots \cdot x_n)]$$

$$= \frac{1}{n} [\log x_1 + \log x_2 + \log x_3 + \dots + \log x_n]$$

$$\Rightarrow \text{GM} = \text{Antilog } \frac{1}{n} [\log x_1 + \log x_2 + \log x_3 + \dots + \log x_n]$$

Example 2:- Consider example 1 and use the logarithm formula to find the GM.

Solution2:- Given 5, 6, 8, 9, 11

Measures of Central Tendency (The Geometric Mean)

$$\text{Thus GM} = \text{Antilog } \frac{1}{5} [\log 5 + \log 6 + \log 8 + \log 9 + \log 11]$$

$$= \text{Antilog } \frac{1}{5} [0.6990 + 0.7782 + 0.9031 + 0.9542 + 1.0414]$$

$$= \text{Antilog } \left[\frac{4.3759}{5} \right] = \text{Antilog } 0.8752 = 7.5021$$

***WHY DO WE NEED A
GEOMETRIC MEAN?????***

Any Questions



Any Questions



Thank
you